



NexGen3D: DIY Steam Machine PRO-V2 - Liquid Cooled BC-250



NexGen-3D-Printing

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Summary

The ultimate BC-250 case.

[Gadgets](#) > [Computers](#)

Tags: [v2](#) [pro](#) [aio](#) [liquidcooling](#) [bazzite](#) [bc250](#)
[steammachine](#) [cachyos](#)

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Update: 11/8/2026:

I have created an assembly guide for the AIO mount here: <https://github.com/NexGen-3D-Printing/SteamMachine/discussions/72>

If you have any questions, you can ask me in that thread.

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Request: 10/8/2026

- **AIO Work:** If anyone is reading this and you know of a common AIO 240 kit that has a pump/block that is 47mm or less, total height,

from cold plate to the pump top, then please send me the details so I can look into it, the Aqua Elite is perfectly fine, but some are having issues buying it, I need to find some alternative's to recommend to people, I can make an alternative mount to support new AIO's, but I cant magically make them lower profile :) I have exactly 48mm from the APU core's face, to the underside of the power supply, the current Aqua Elite, is 47mm's in height.

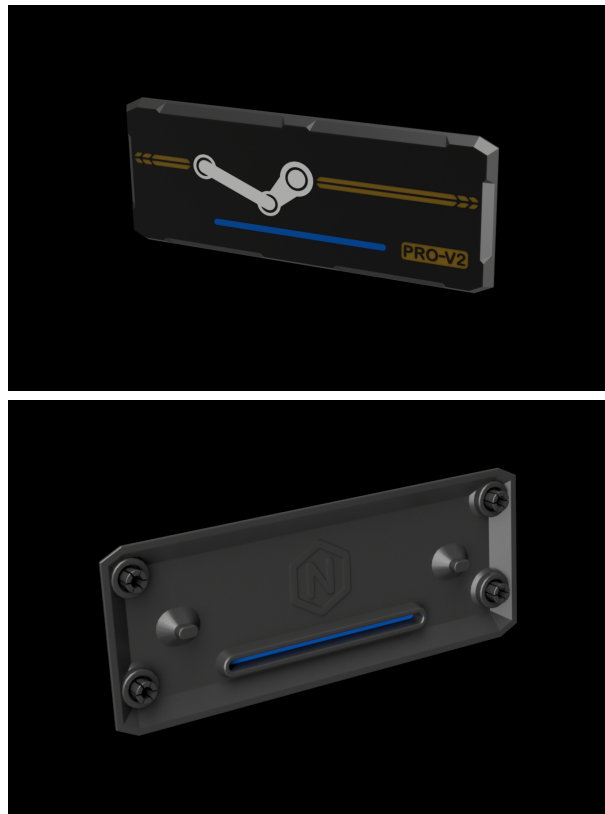
- **Software Improvements:** Im also looking for someone to help roll all of my settings, tweaks and setup scripts into one nice interactive tool for Bazzite, if you think this is something you can do, or help with, please contact me.
- **Forum:** If you want to follow my design updates, as most things getting posted there first, please head over to my Github repo discussions area: <https://github.com/NexGen-3D-Printing/SteamMachine/discussions>

Update: 9/08/2026

- I have now uploaded all 4 Version 2.1 Face Plates, I recommend using these over the previous version as they are much more durable, and I think they look better, but I have left the old ones up in case you prefer the old style instead.
- I've also uploaded a blank STEP file of the face plate for those that would like to create their own custom design, or incorporate their own screen or ARGB setup.
- Due to the complex setup of the TURZX 5.2" display, I have now ordered a 6.8" screen from Thermalright, these seem to have far better Linux support as there is at least 3 current projects I have found for them, and they appear to have full GUI setup, so I will create a 5th Faceplate option for this wide screen style display.

Update: 6/08/2026

- I have uploaded 2 new Face Plates, these are version 2.1
- I have beefed up the locking studs now, and I have added guid pins to resist the stud tabs from breaking when lateral force is applied to the face plate:



Update: 31/07/2026

Online Store: For those looking to purchase my cases, I'm not working on a deal with DHL for international shipping, so I should be able to start shipping to the USA and all other regions as well.

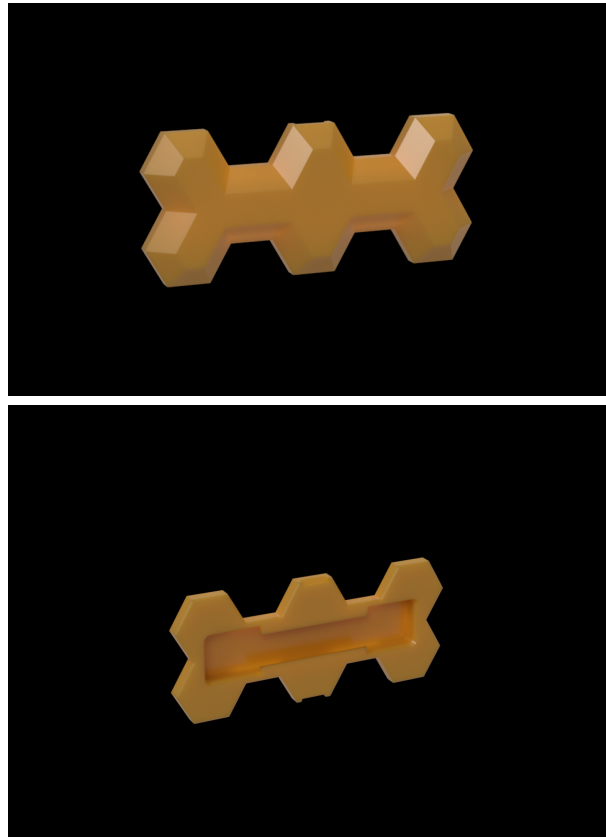
Keep your eye on my store, as it will be getting updated over the next few days, to reflect the updated designs, new colour schemes, plug and play cables, and with stock ready to ship: <https://nexgen3d.bigcartel.com>

Update: 30/07/2026

- Vertical Mode has been Enabled
- Just pop the power button out and rotate 90 degrees and pop it back in, then install the feet and now you have a mini tower beast.
- The feet are designed to be printed from PETG and TPU and just slot into the Hex pattern like the long foot for the horizontal mode, I also have an alternative vertical foot that can be printed in just PETG, and installs using a single M3 12mm bolt in each.
- It is critical that you do not stand this unit on its side without the feet, or you will inhibited the intake air flow, these feet are designed to maintain proper fresh air intake, which is critical to keep all the nasty hot points nice and cool.
- The vertical mode really sets this design off, it looks awesome in person.



- I've also added a blank Rune Stone Emblem that is designed to accept some ARGB strip light, best to print this from a translucent colour, there is a small hole in the lefthand side access panel to allow the cables to pass through.



To all that have been waiting, I'm sorry on the delay, but I have been super busy, not enough time in the day for everything.

I'm proud to introduce the all new DIY Steam Machine PRO-V2, just like my REDUX, this new case has many improvements and bug fixes over the previous version.



Please Note: This page will be receiving updates daily as there is a few things I still need to link to and add.

To list a few of the changes:

- Optimised design for 0.6mm nozzle printing (0.4mm is fully support as well)
- The one case now supports Flex-ATX (Includes the FSP500), Meanwell LOP and Server PSU
- Mounting location for the Mosfet Party Lite Power PCB.
- Added cable management points
- Backlit Power Button
- 5.2" TURZX Stats Display Option
- Improved VRAM cooling
- Improved front and rear case section interlocking with screw options for those that want to run without the hex panels.
- Lefthand side access panel for easier assembly
- Stronger hex panels
- Internal multipurpose bay, for additional items like a USB hub, SATA or USB drives, or even an ESP BT Power on Board, or a combination of all.
- Optimised internal geometry for easier printing
- Better, more durable removeable face plate design
- Multiple hex side panel options.
- Improved AIO mount

All the hardware from the original case is compatible with this new design, with the exceptions of you needed 2 x M3 x 8mm pan head bolts, these are needed for securing the power button.

If you would like to make a donation or sponser my ongoing work to the open community, here is my: Ko-Fi

If you would like to help with creating some software specific for this project, please contact me in the discussions area on my [Github Repo](#)

AIO Bill Of Materials:

- 4 x [M3 x 50mm Bolts](#)
- 12 x [M3 x 14mm Bolts](#)
- 24 x [Flanged M3 Threaded Inserts \(3.5x4.6x6\)](#)
- 4 x [0.9mm Wire, 6mm outer diameter x 10mm long Springs](#)

Heatsink Kit for the VRM's:

- The easiest kit is this one, has more than you need, but its cheap: [GeekPi 102 Piece Heat Sink Kit](#)
- I also recommend picking up some thermal plaster, as sometimes the sticky pads on the heatsinks don't work, if that's the case, just glue them on with this stuff: [Thermal Plaster](#)

- Another option some have done, is cutting the stock heatsink and putting that on, if you do, just remember to open the fin stack up, but leave the outer few rows closed, to help hold it together, as it tends to fall apart.

Recommended Material:

- ABS or ASA, Glass Fibre or Carbon Fibre are better
- **PET-CF** (This is not PETG-CF, don't confuse them, they are different)

Case Bill Of Materials:

- Fasteners:
 - 29 x **Flanged M3 Threaded Inserts (3.5x4.6x6)**
 - 29 x **M3 x 12mm Bolts** (Can use 10mm to 14mm, but 12mm is recommended)
 - 2 x **M3 x 8mm Wafer Thin Head Bolts**
- Electronics
 - **Latching Power Button** (Used with Flex-ATX and FSP500 connected directly to the PSU) Needs to be Latching, 12v and 16mm you will also need the **wire loom**
 - Optional but highly recommended, **Momentary Power Button** and **Mosfet Party Lite PCB** switch needs to be Latching, 12v and 16mm you will also need the **wire loom**
 - Recommended Power Supplies:
 - **FSP500-30AS** <- This PSU is GOAT'ed just off the guy \$10 to \$15 USD for it
 - Metalfish 550M <- Avoid this one at the moment, the Sharks on Aliexpress are on drugs, they must think the 80+ Gold means its packed full of Gold.
 - Liquid Cooling Kit - Thermalright Aqua Elite 240
 - **Amazon Australia**
 - **Amazon USA**
 - **Arctic P12 Pro Fans**
 - **Arctic P8 Fan**
 - **Arctic MX-7**
 - **Corsair Commander Duo**
 - **8 Pin PCI-E to Sata Power Cable** (Needed only for the Corsair Hub and the FSP500-30AS)
 - ARGB Strip:
 - If you want to make your own high density one, then go **here**
 - If you just want a pre made one, then **here**
 - **Anker USB 3 Hub**, dont look for alternatives, as they wont fit, you can however, drop a second hub inside the case, there is plenty of room under the radiator.
 - Optional **TURZX 5.2" Stats Display**

- **2 x 180 Degree USB Adaptors** (You need the Black one, do not buy the blue one, it won't fit, see image below)



I also have this Micro-Fit locking connector for the BC-250, this makes it safe so you don't mix up the plugs and let the magic smoke out of the board: <https://www.printables.com/model/1785063-nexgen3d-micro-fit-bmi-retainer-for-the-bc-250>

Recommended Materials:

- **PETG with Carbon Fibre or and PETG with Glass Fibre**

Parts To Print:

- AIO Mounting Bracket
 - 1 x Thermalright Mount Top - V2
 - 1 x Thermalright Mount Bottom - V2
 - 1 x Clamping Plate - V2
 - 4 x Thumbnut - Mount Tensioner - V2
 - 4 x Thumbnut - Back Plate Retainer - V2
 - 8 x Spring Seat - V2
- Case
 - TBA

Alternative Power Supply Options:

- LOP-400/500/600
- Server Power Supply

Video Build Guide: -> In Production <-

Software Install Video Guide: -> In Production <-

Links:

- **My Github Repo**
- **All About the BC-250**
- **Technical Information**
- **Old Lamer's YouTube Channel**
- **My YouTube Channel**

- [My Website](#)
- [My Ko-Fi](#)

Model files

 **Latest Face Plate Design - (Version 2.1)** 5 files

	faceplate-option-1.3mf
	faceplate-option-2.3mf
	faceplate-option-3.3mf
	faceplate-option-4.3mf
	blank-face-plate.step

 **Vertical Stand** 3 files

	vertical-stand-optio-1-petg.3mf
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vertical-foot-option-2.3mf



vertical-stand-optio-1-tpu.3mf



Power Supply Mounts

4 files



pro-v2-meanwell-lop-backplate-400-500-600.3mf



pro-v2-meanwell-lop-mount-400-500-600.3mf



pro-v2-server-psu-mount-hp-dell-lenovo.3mf



pro-v2-flex-mount-fsp500-and-flex-atx.3mf



Power Button

3 files



pro-v2-transparent-rear-section.3mf

☐ Print from Transparent Clear or a Translucent White



pro-v2-steam-logo.3mf



pro-v2-button-mounting-plate.3mf

☐ Requires 2 x M3 x 8mm Pan Head Bolts



Top and Bottom Hex Panels

2 files



pro-v2-top-hex-panel.3mf



pro-v2-bottom-hex-panel.3mf



Right Hand Hex Panel

5 files



pro-v2-rh-hex-panel-logo-edition.3mf



pro-v2-rh-hex-panel-rune-stone-edition.3mf



pro-v2-rh-hex-panel-no-logo-edition.3mf



rune-stone-logo.3mf



rune-stone-argb.3mf



Left Hand Hex Panel

5 files



pro-v2-access-panel-required.3mf



pro-v2-lh-hex-panel-logo-edition.3mf



pro-v2-lh-hex-panel-no-logo-edition.3mf



pro-v2-lh-hex-panel-rune-stone-logo-edition.3mf



rune-stone-logo.3mf



Foot

2 files



pro-long-foot-petg.3mf



pro-long-foot-tpu.3mf



Face Plates (Old - Version 2)

4 files



faceplate-option-1.3mf



faceplate-option-2.3mf



faceplate-option-3.3mf



faceplate-option-4.3mf



ARGB LED Strip

2 files



pro-v2-argb-strip-mount.3mf



pro-v2-diffuser.3mf



Case

3 files



pro-v2-rear-case-multi-material.3mf



pro-v2-front-case.3mf



pro-v2-usb-cover-multi-material.3mf



AIO Mount

6 files



thumbnut-mount-tensioner-v2.3mf



thermalright-mount-bottom-v2.3mf



clamping-plate-v2.3mf



thermalright-mount-top-v2.3mf



thumbnut-back-plate-retainer-v2.3mf



spring-seat-v2.3mf

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